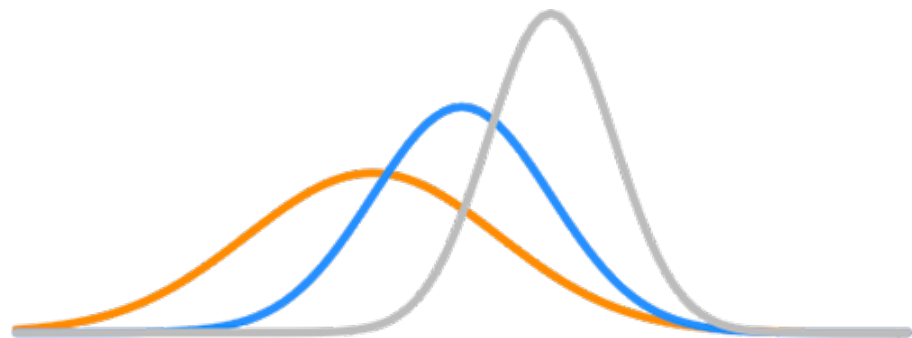
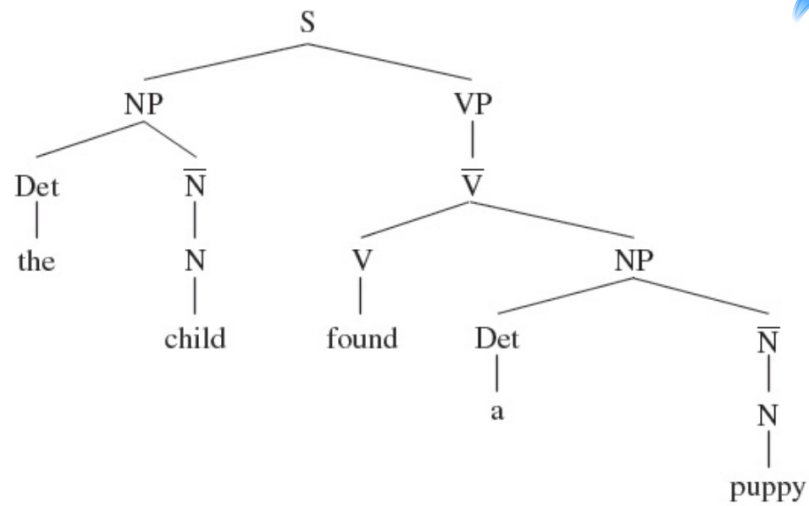


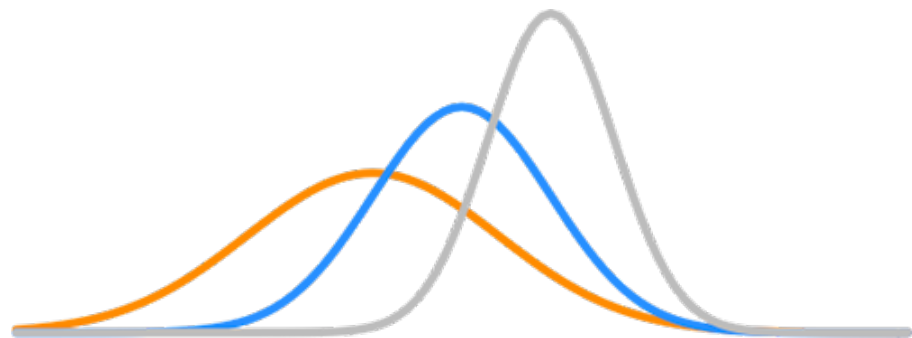
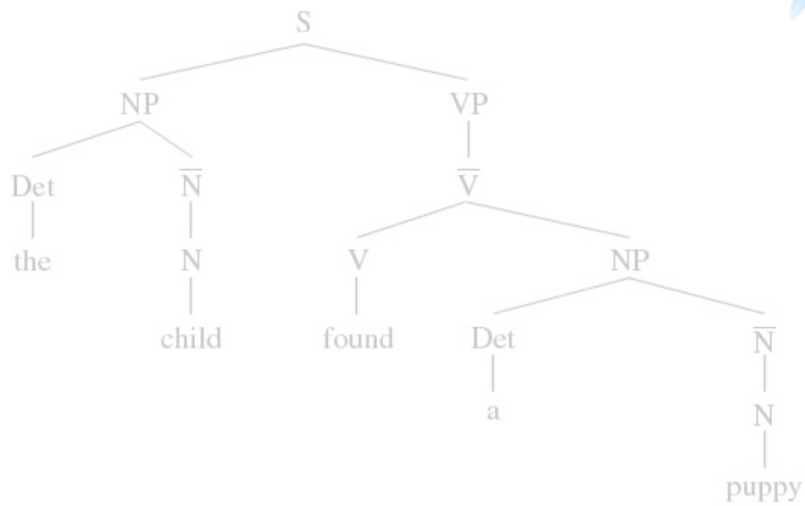
Numerical optimization

Computational modeling of language
Spring 2026

Bureaucracy

- **None today!**





Our lineup for today

- Marr recap
- Language universals and numerical optimization
- Liljencrants and Lindblom

Our lineup for today

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- Liljencrants and Lindblom

Marr's (1982) levels of analysis

Marr's three levels

The three levels at which any machine carrying out an information-processing task must be understood

*Computational
theory*

*Representation and
algorithm*

*Hardware
implementation*

Marr's three levels

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What is the goal
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why is it
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How can the representation and algorithm be realized physically?

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The functional goal of the visual system is to determine the shape of objects in the world.

Marr's three levels

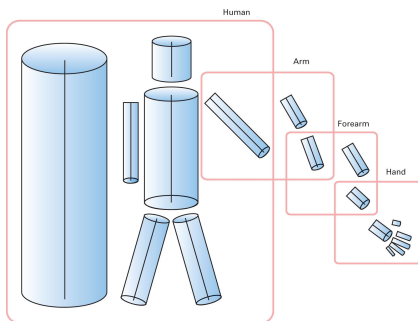
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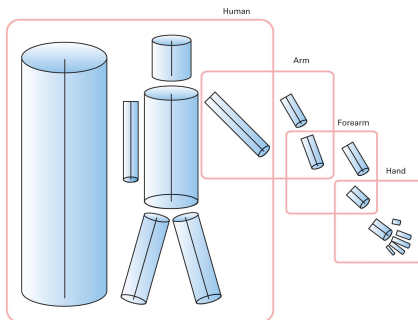
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Neurons,
somehow.

Marr's three levels

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The goal is to
fly.

Curved wings,
aerodynamics

Feathers, balsa
wood, steel,
other...

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What about language?

The computational level

Specifies “what the system is up to” – what the **functional goal** of its processing is.

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Allows us to ask: what is the **optimal** solution for that goal?

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Specifies “what the system is up to” – what the **functional goal** of its processing is.

Allows us to ask: what is the **optimal** solution for that goal?

And do humans do it that way – regardless of algorithmic and implementational detail?

Our lineup for today

- Marr recap
- Language universals and numerical optimization
- Liljencrants and Lindblom

Constrained variation

Constrained variation

Some examples

(Evans & Levinson 2009)

Constrained variation

Sound inventories: Number of phonemes ranges from 11 to 144.
“[W]e do not want to think about universal phonetic **categories**, but rather about universal phonetic **resources**, which are organized and harnessed by the cognitive system”
(J. Pierrehumbert)



Janet Pierrehumbert
Oxford University

Constrained variation

Syllables and the CV universal:

- Proposal: **CV > V > VC** – i.e. a language will have V syllables only if it also has CV syllables, and will have VC syllables only if it also has V syllables (Jakobson and Halle 1956).

Constrained variation

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- **Falsified** by Arrernte, which has VC without CV.
- Generalization is still true as **tendency**, but not absolute.

Constrained variation

Morphology:

Continuum from

- **Isolating** languages: tend to lack bound morphemes such as affixes (e.g. cat-s), to

Constrained variation

Morphology:

Continuum from

- **Isolating** languages: tend to lack bound morphemes such as affixes (e.g. cat-s), to
- **Polysynthetic** languages: an entire English sentence can sometimes appear as a single word: e.g. “I will plant potatoes for them again” (Cayuga).

Constrained variation

Semantics:

- There **cannot** be a **universal repertoire of concepts** that map onto linguistic forms in different languages, because languages differ substantially in their semantic repertoires.

Constrained variation

Semantics:

- There **cannot** be a **universal repertoire of concepts** that map onto linguistic forms in different languages, because languages differ substantially in their semantic repertoires.
- E.g. in some languages, one cannot say “**let’s go**” without specifying whether **by foot or by vehicle**. Many other examples.

Constrained variation

Recursion: Some languages **show limits on recursion**, or lack it altogether.

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- The doubly-embedded English sentence:
“[They stood [watching us [fight]]]”

Constrained variation

Recursion: Some languages **show limits on recursion**, or lack it altogether.

- The doubly-embedded English sentence:
“[They stood [watching us [fight]]]”
- is expressed in Bininj Gun-wok **without embedding**, as:
“they_stood / they_were_watching_us /
we_were_fighting_each_other”
- where _ links morphemes within a word.

Instead of strict universals

- Universal **tendencies: clusters plus outliers**

Instead of strict universals

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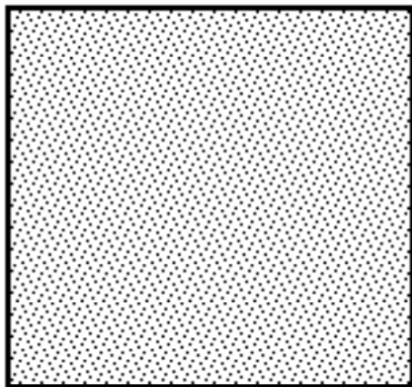


Fig. 1b.

Instead of strict universals

- Universal **tendencies: clusters plus outliers**

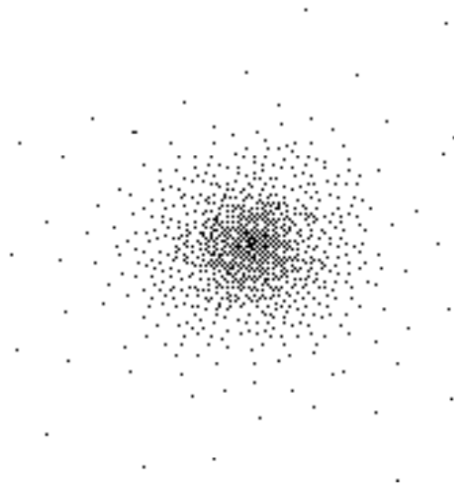


Fig. 1a

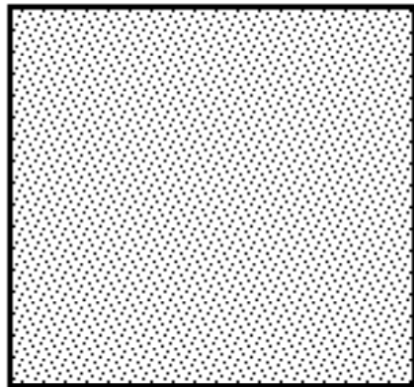


Fig. 1b.

Instead of strict universals

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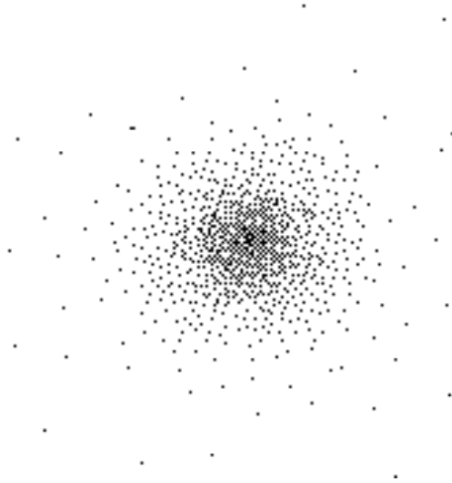


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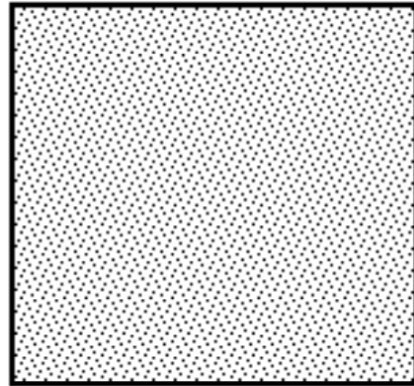


Fig. 1b.



Why statistical universals
are better than absolute
universals

Matthew Dryer, 1998

Instead of strict universals

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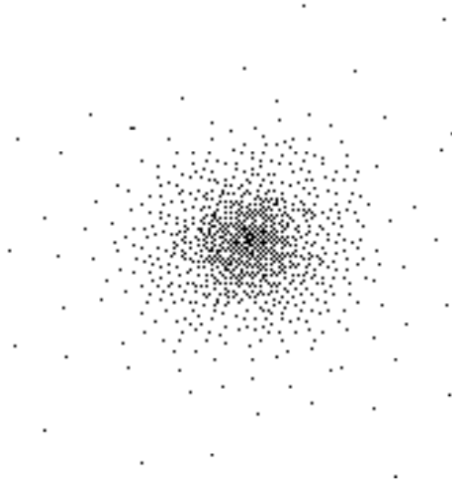


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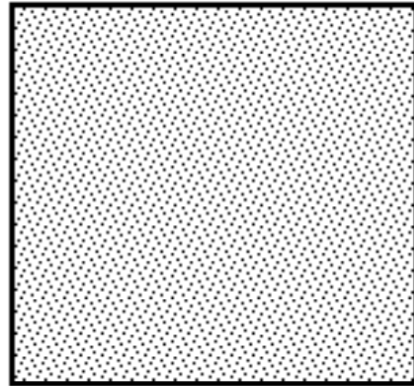


Fig. 1b.



Why statistical universals
are better than absolute
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Matthew Dryer, 1998

“Making a **virtue** out of a
necessity” - E&L

Instead of strict universals

- Universal **tendencies: clusters plus outliers**
- Many languages mark the **negative** by introducing a morpheme into the sentence (e.g. English “**not**”).

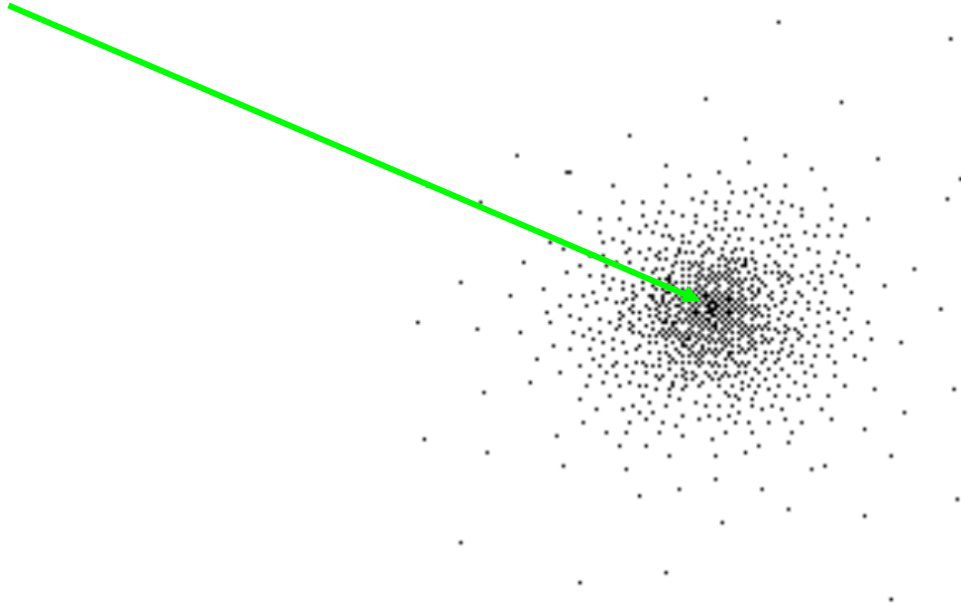
Instead of strict universals

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- But classical Tamil instead marks the negative by **deleting tense morphemes** that are present in the corresponding positive form of the sentence.

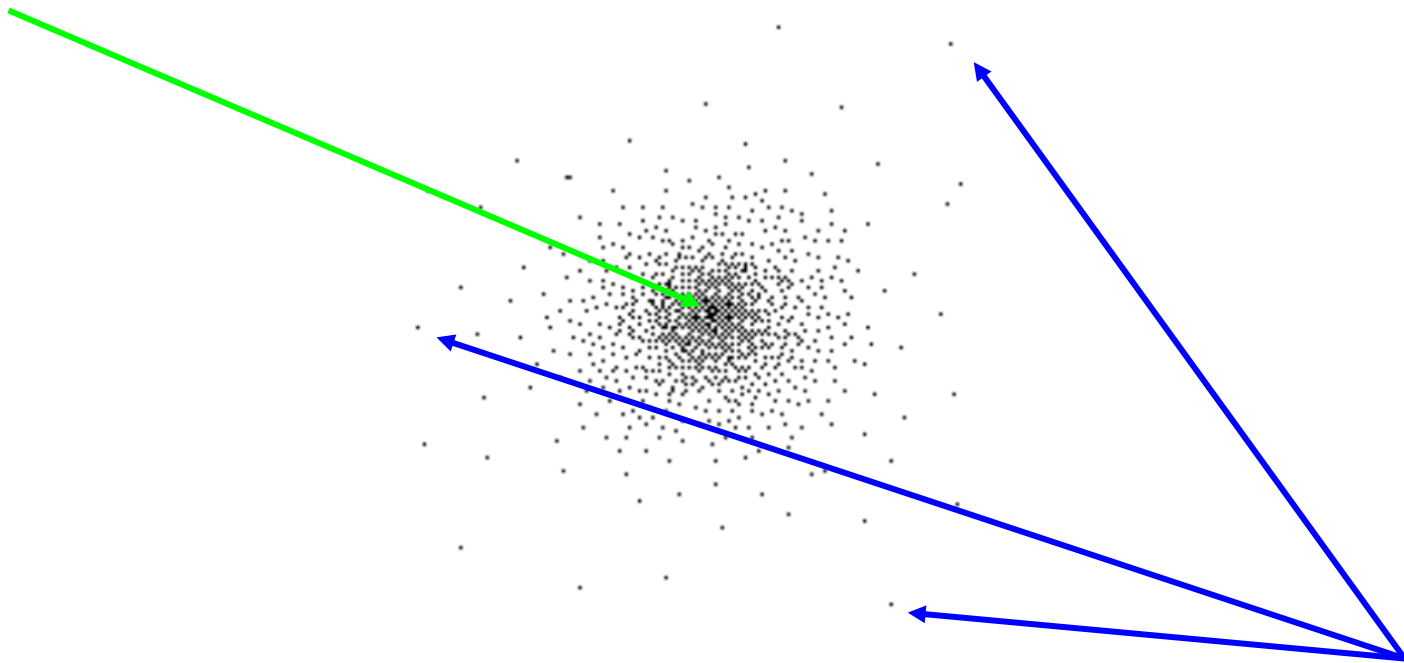
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- But classical Tamil instead marks the negative by **deleting tense morphemes** that are present in the corresponding positive form of the sentence.
- See [rarum 33](#) in Frans Plank’s [Raritätenkabinett](#).

Account **both** for the
cluster center ...



Account **both** for the
cluster center ...



... and for the **outliers**.

How?

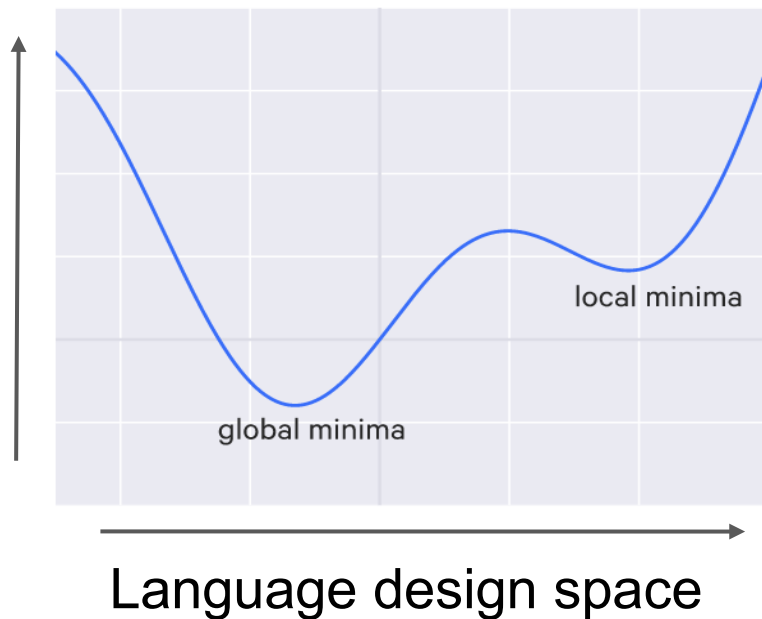
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- Human languages as evolutionarily stable strategies, or **local optima**, in a landscape of possible languages.

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Communication cost



How?

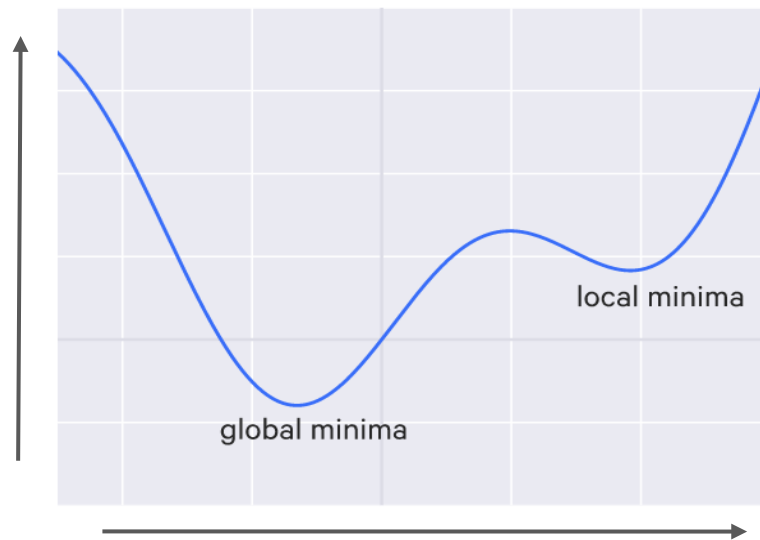
- Human languages as evolutionarily stable strategies, or **local optima**, in a landscape of possible languages.
- **Recurrent** but **not universal** solutions to the problem represented by the landscape: how to **communicate** about just about anything.

How?

- Human languages as evolutionarily stable strategies, or **local optima**, in a landscape of possible languages.
- **Recurrent** but **not universal** solutions to the problem represented by the landscape: how to **communicate** about just about anything.
- Accommodates the **cluster-plus-outliers** nature of the data.

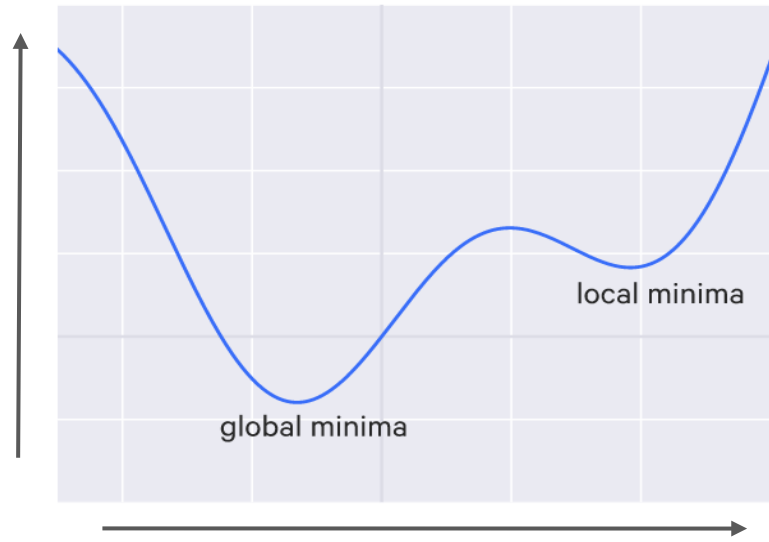
OK, but how?

Communication cost



Language design space

$f(x)$

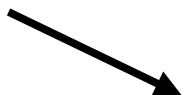


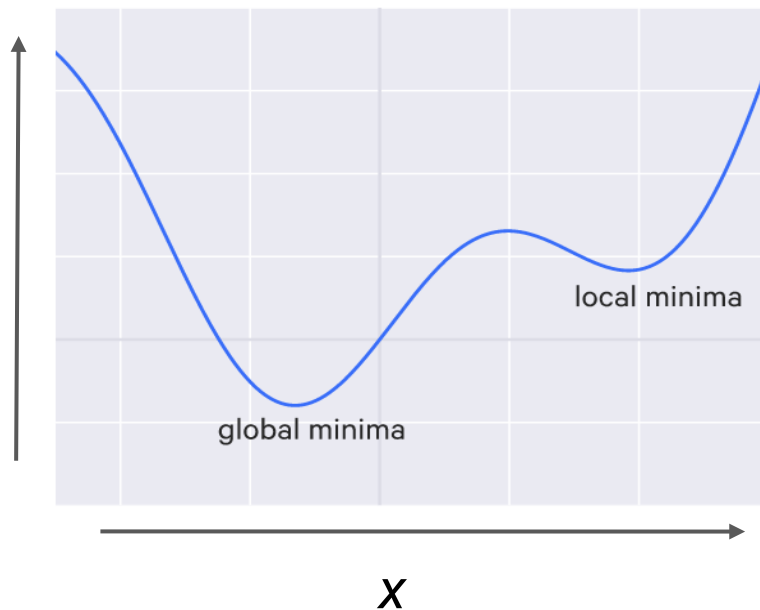
global minima

local minima

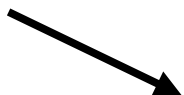
x

**Objective function,
or loss function**

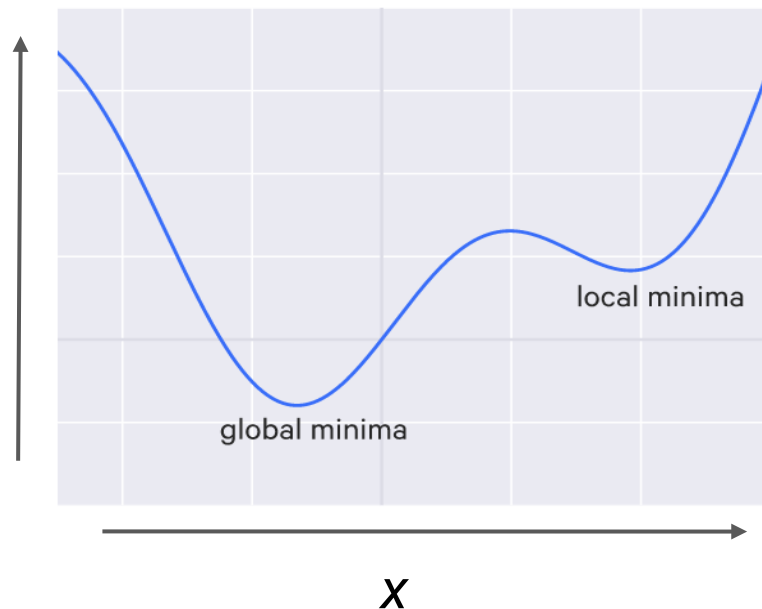
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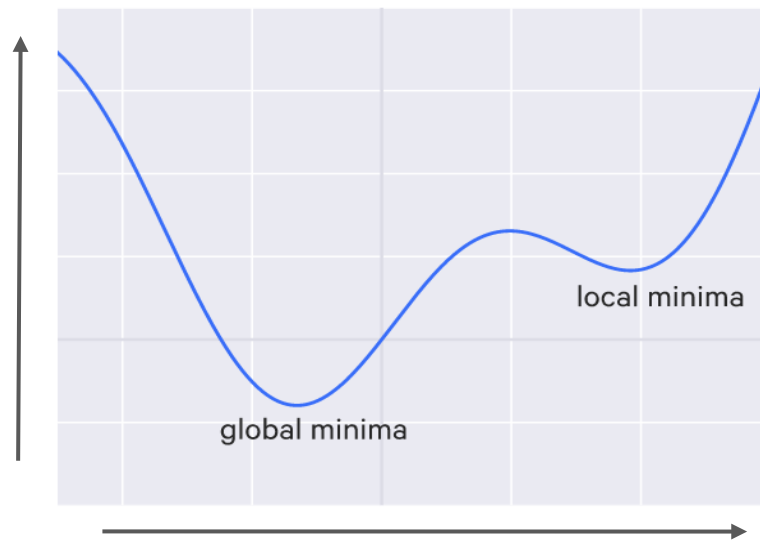
 $f(x)$

We seek $\arg \min_x f(x)$



An interactive tutorial on numerical optimization

Communication cost



Language design space

Our lineup for today

- Marr recap
- Language universals and numerical optimization
- Liljencrants and Lindblom

NUMERICAL SIMULATION OF VOWEL QUALITY SYSTEMS: THE ROLE OF PERCEPTUAL CONTRAST

JOHAN LILJENCANTS and BJÖRN LINDBLOM

Royal Institute of Technology, Stockholm

A numerical model is developed in order to establish the extent to which the principle of maximal perceptual contrast can be used in phonological theory to explain the phonetic structure of vowel systems.

Language, Vol. 48, No. 4 (Dec., 1972), pp. 839-862

Liljencrants & Lindblom 1972

- Seeks to **explain cross-language variation** in vowel inventory, in **phonetic** terms.

Liljencrants & Lindblom 1972

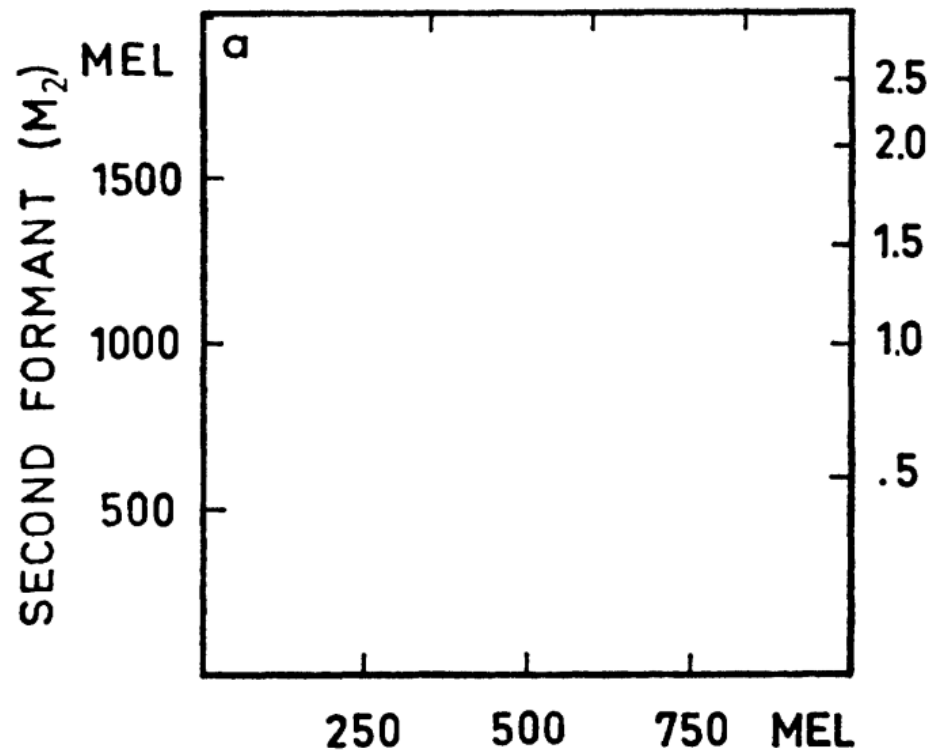
- Seeks to **explain cross-language variation** in vowel inventory, in **phonetic** terms.
- Grounds **discrete symbols** in a **numerical continuum**, itself ultimately grounded in **anatomy** and **communicative** principles.

Liljencrants & Lindblom 1972

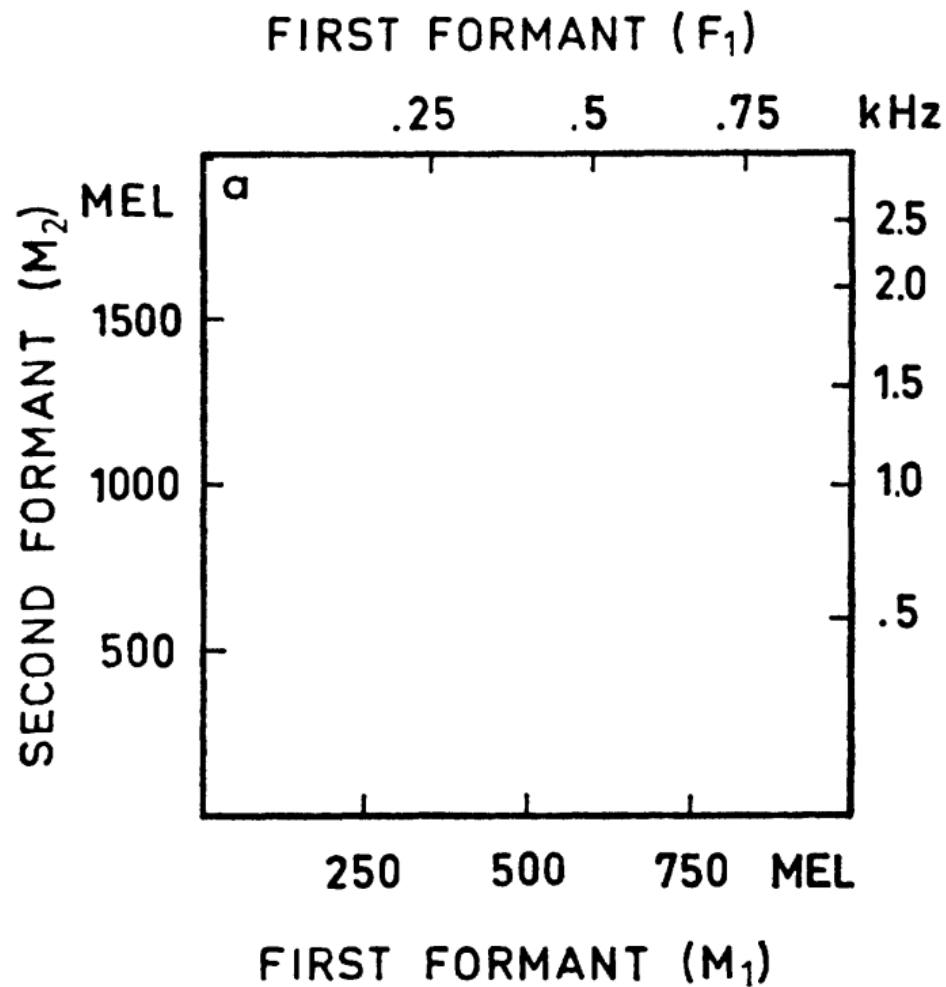
- Seeks to **explain cross-language variation** in vowel inventory, in **phonetic** terms.
- Grounds **discrete symbols** in a **numerical continuum**, itself ultimately grounded in **anatomy** and **communicative** principles.
- Specifically: principle of **maximal perceptual contrast**, for maximum **discriminability**.

FIRST FORMANT (F_1)

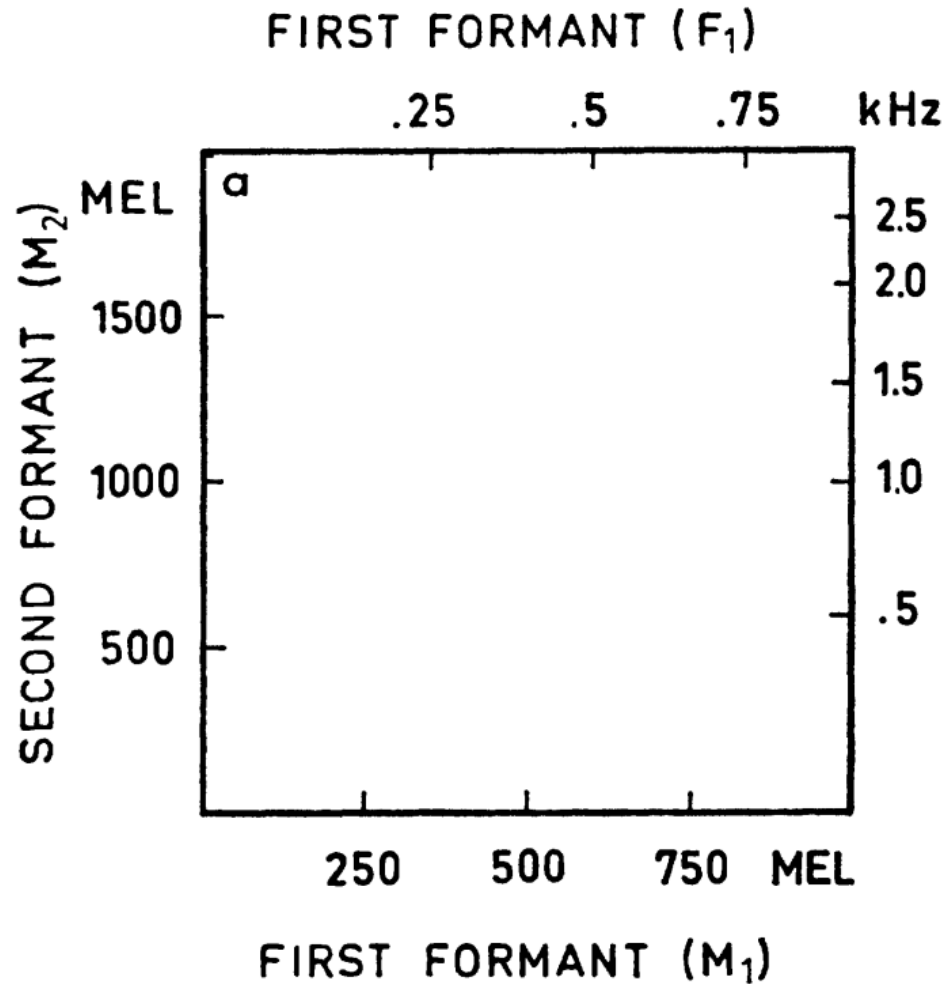
.25 .5 .75 kHz



FIRST FORMANT (M_1)

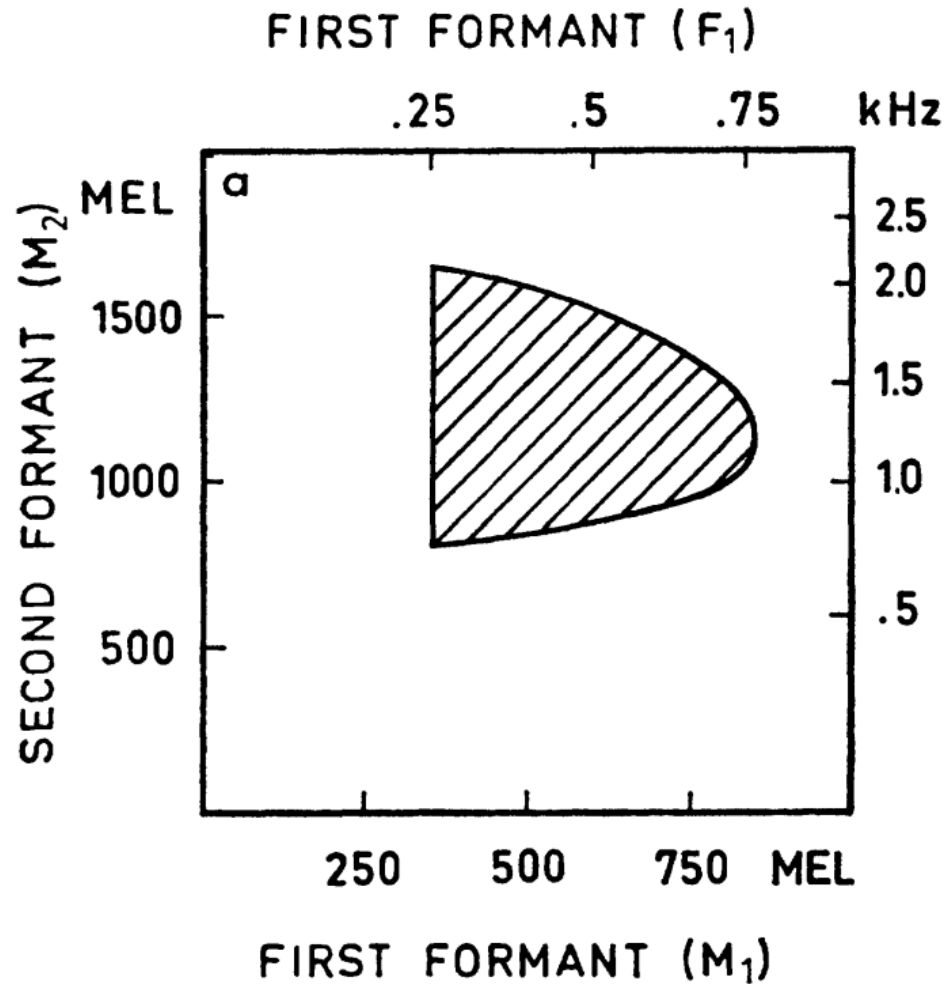


Formant: a concentration of acoustic energy around a particular frequency in the speech wave.



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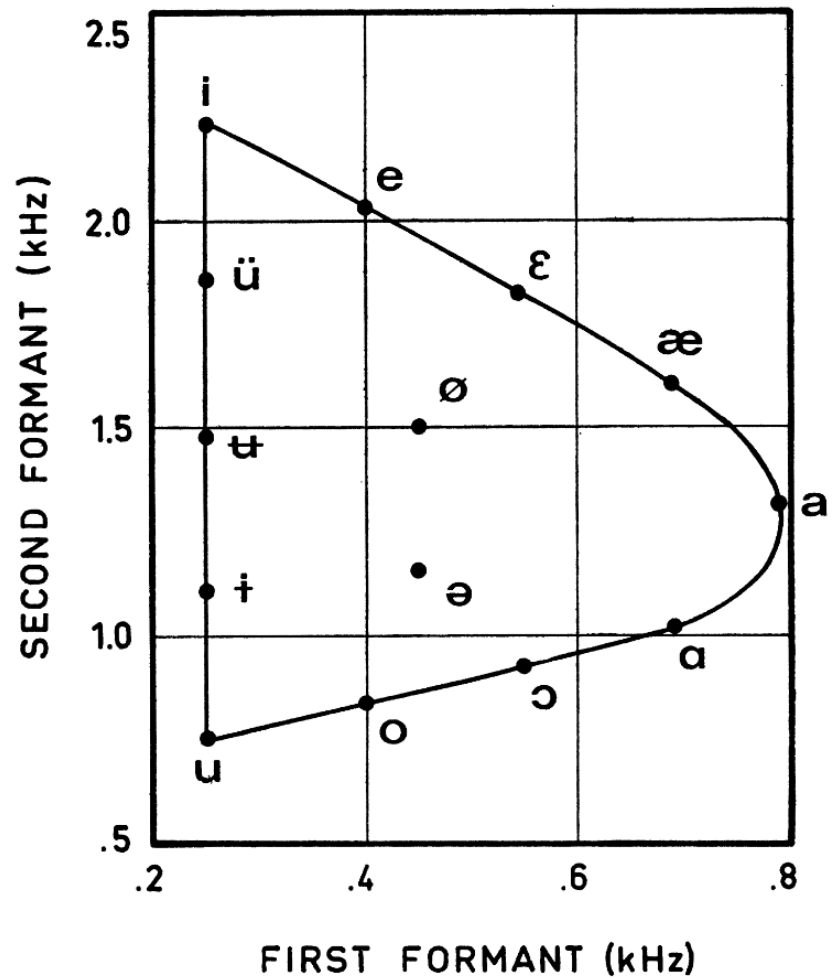
The **mel** scale: a perceptual scale of pitches judged by listeners to be equal in distance from one another.

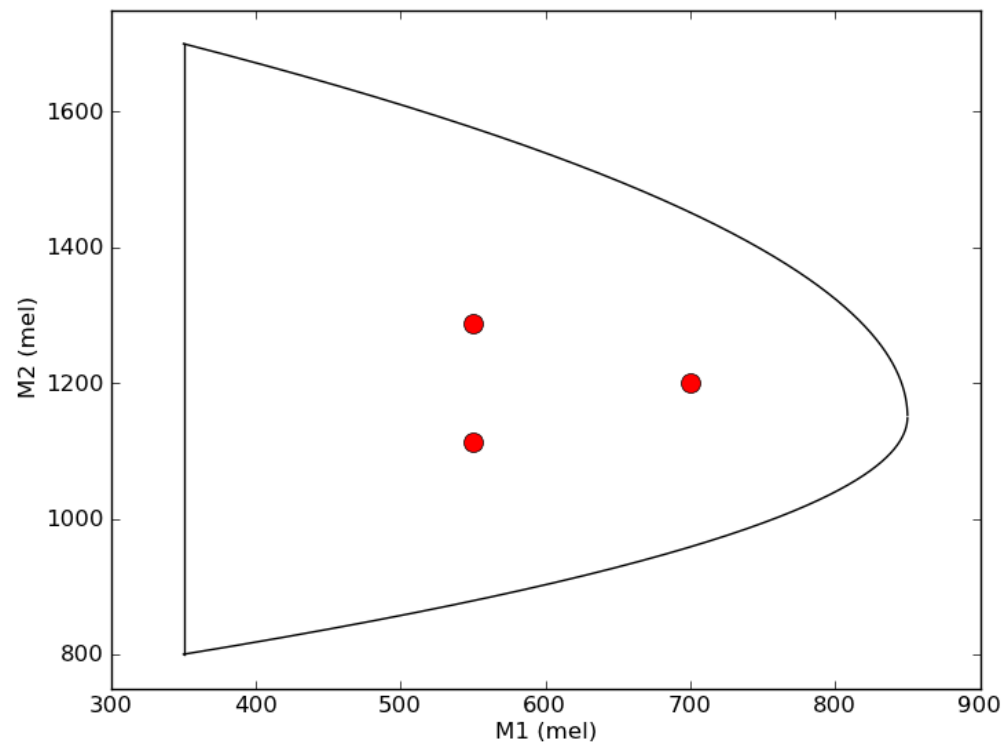


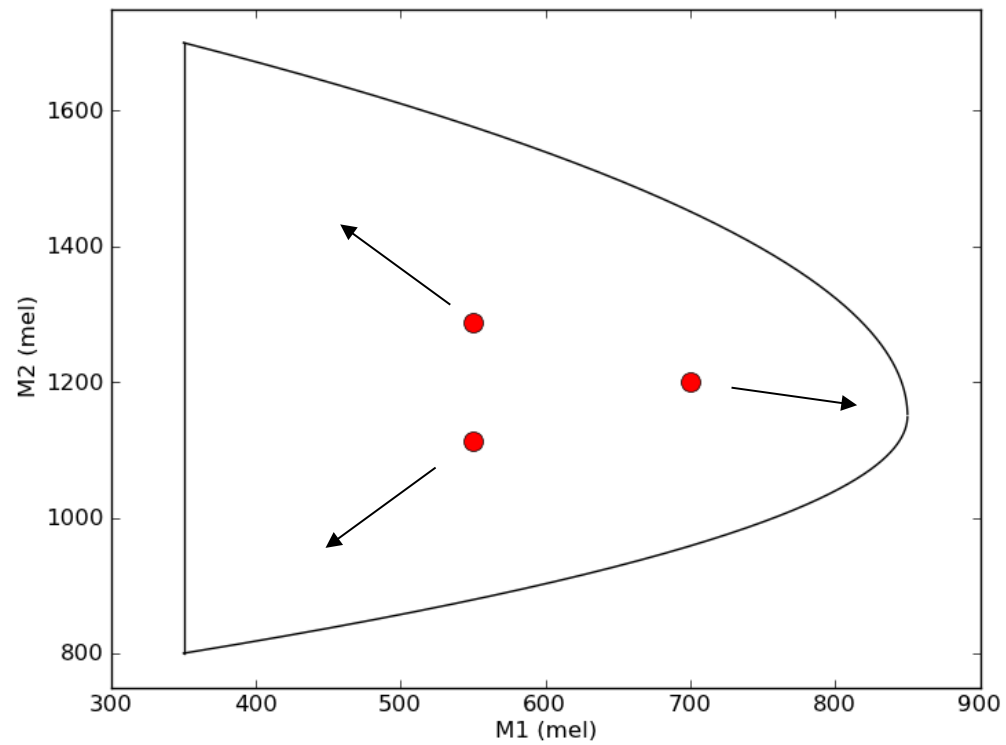
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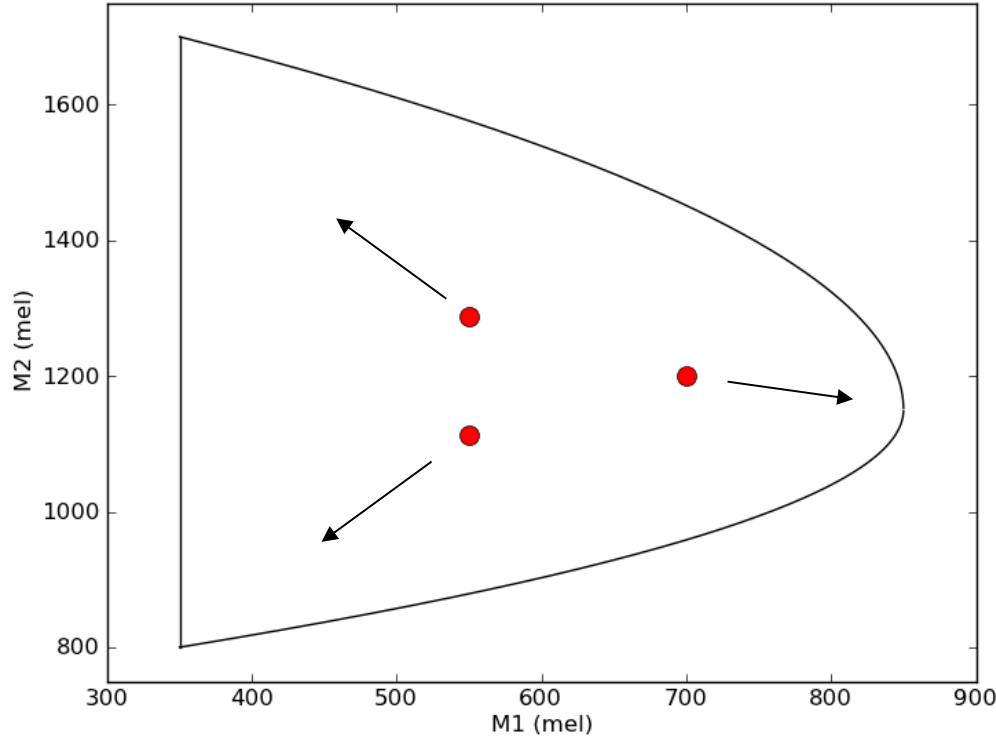
The **mel** scale: a perceptual scale of pitches judged by listeners to be equal in distance from one another.

Shaded region: the range of articulatorily possible human vowel sounds.

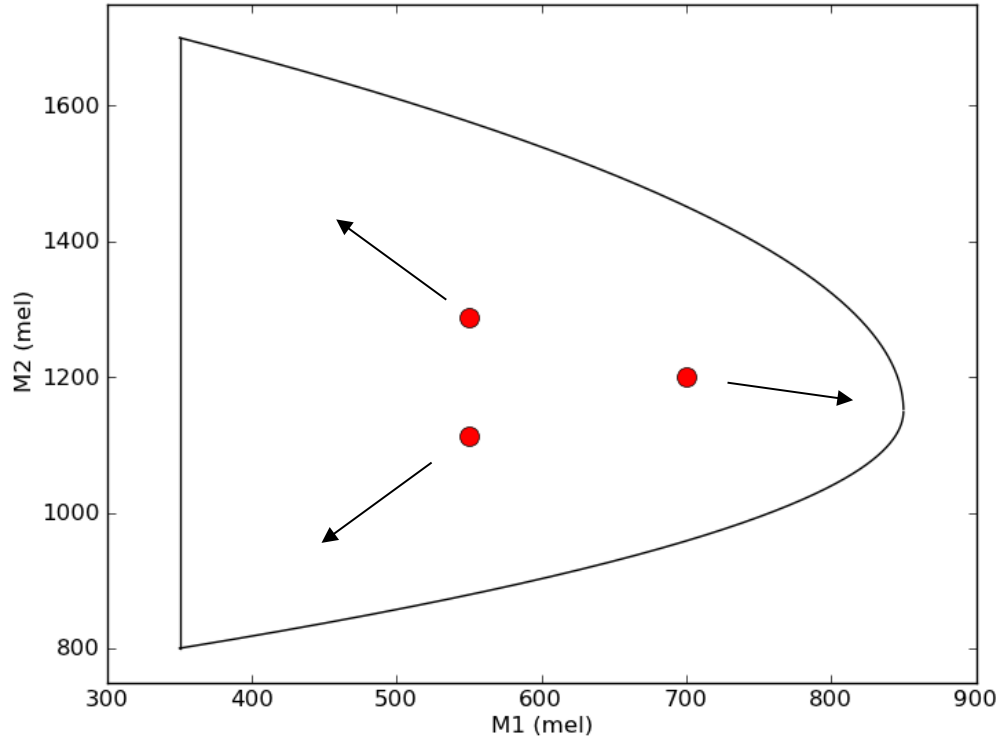






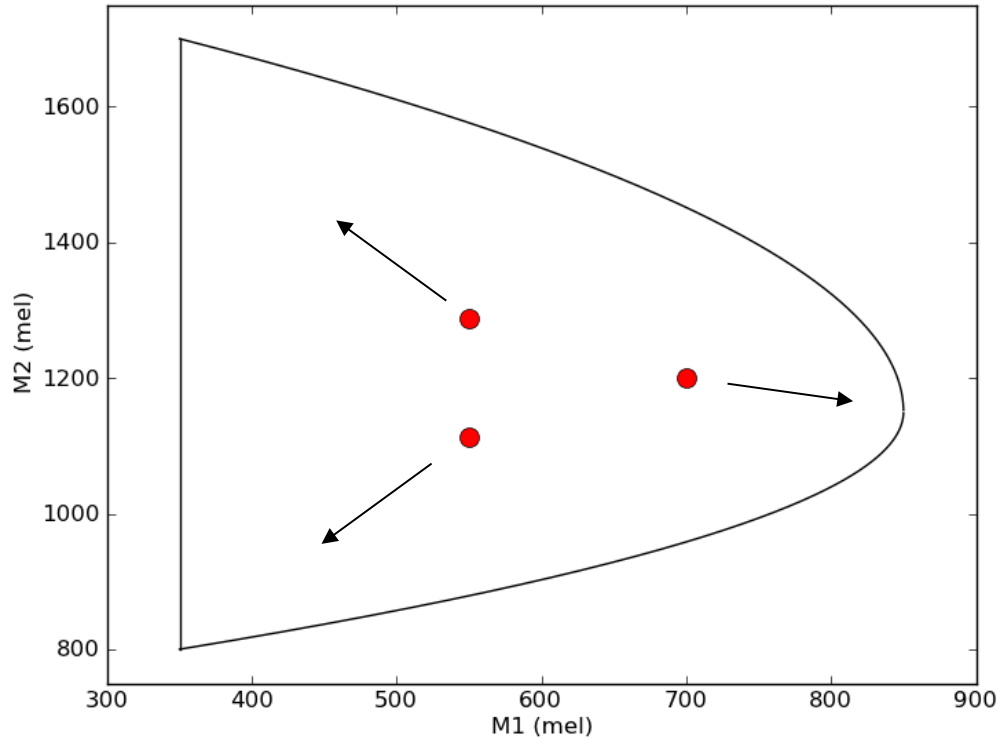


Intuition: physical particles repelling each other, and moving in an enclosed physical space to be as far from each other as possible.



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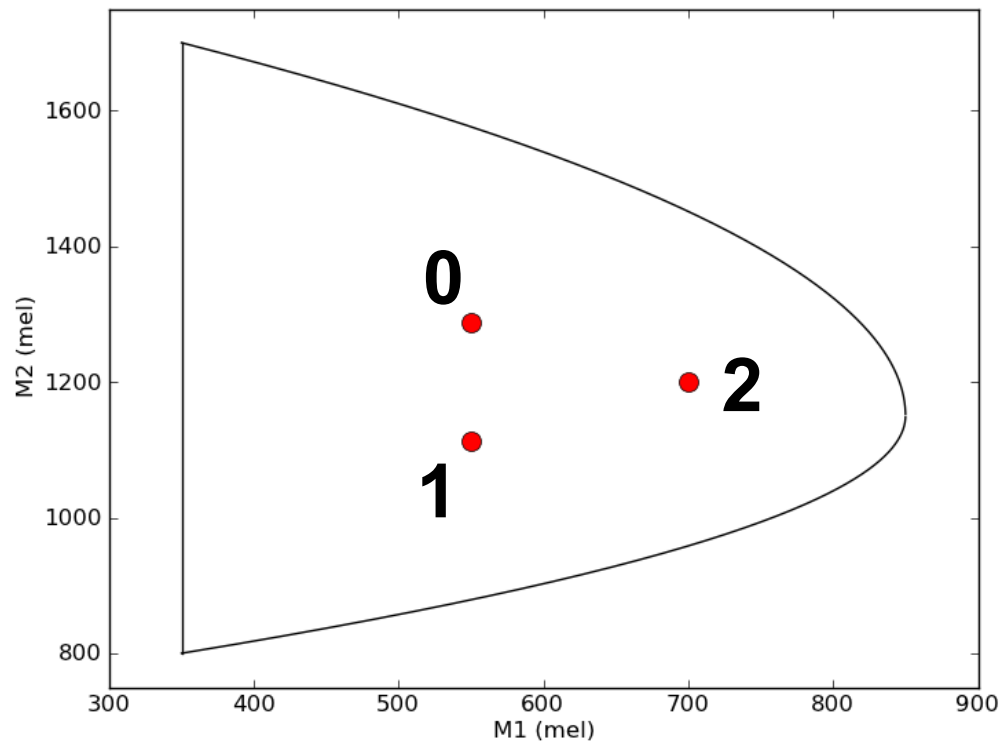
E.g. **cork-mounted magnets** on a water surface bounded by some shape.

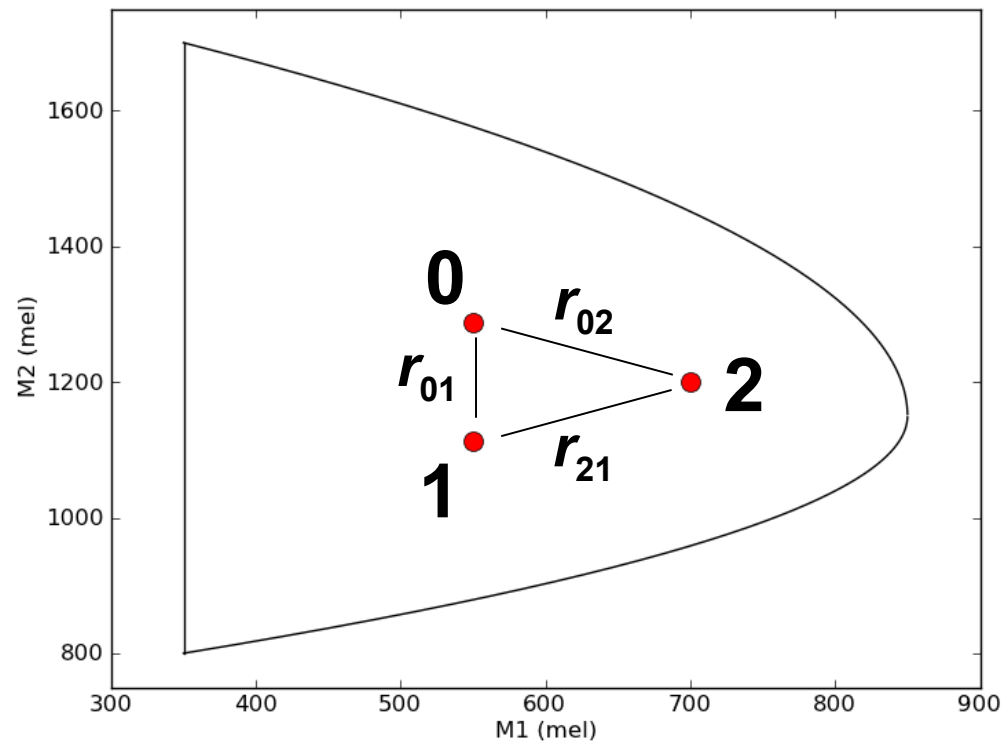


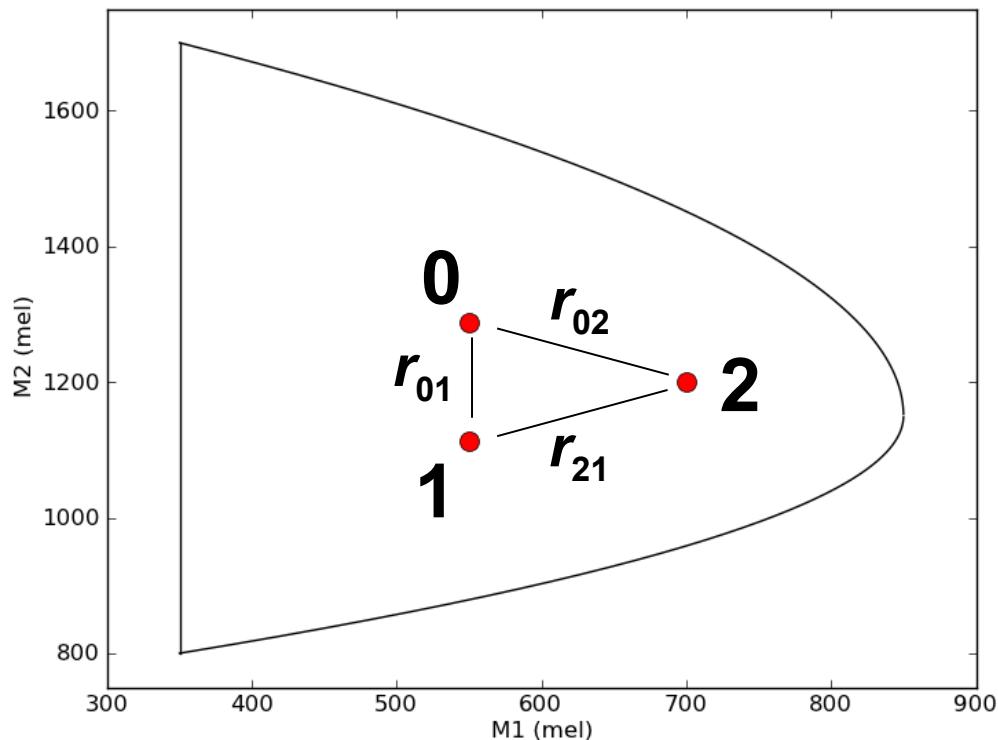
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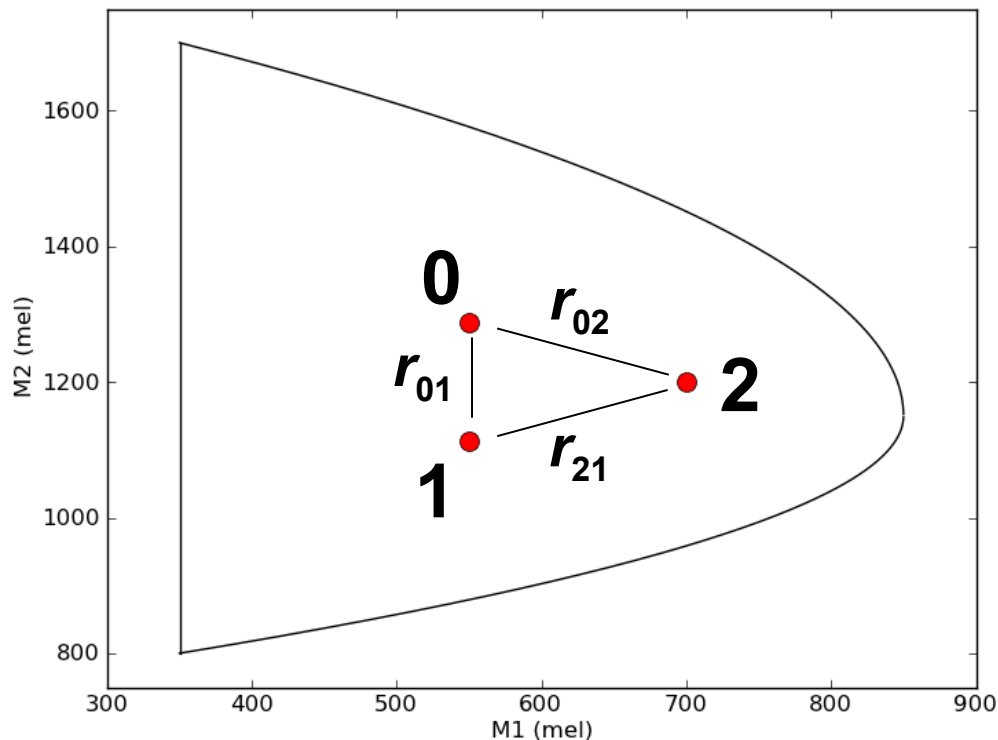
They do the same, with the same equation, for **vowels**.







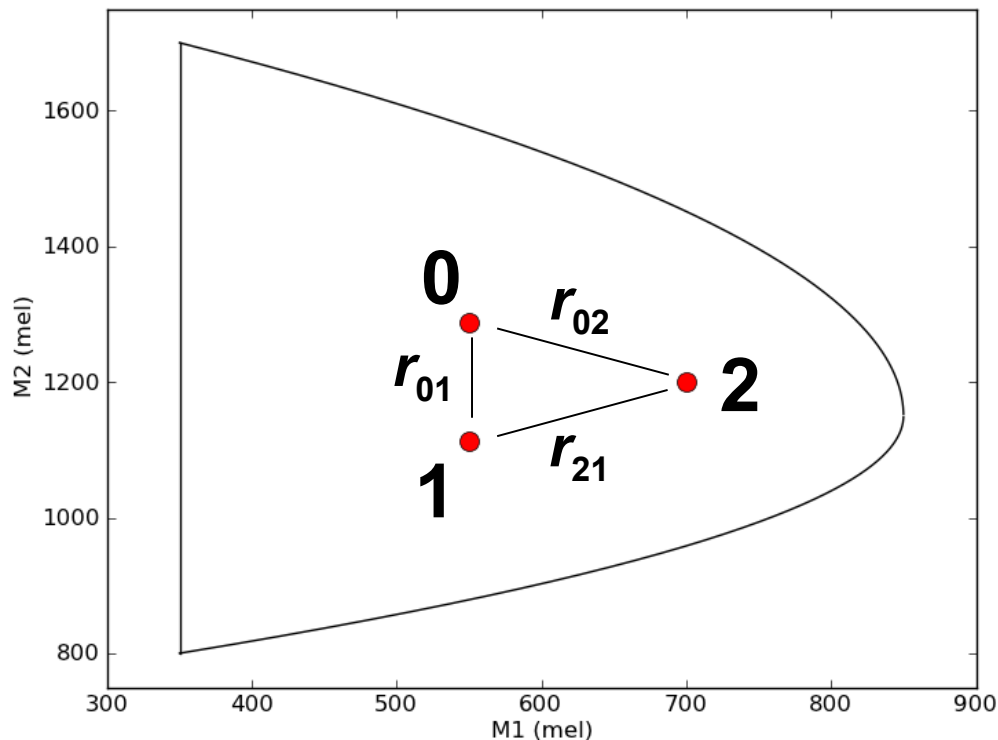
They seek vowel coordinates x_i, y_i , for all i , that **minimize** the energy **E** of a vowel system, where that quantity is borrowed from physics and obeys an inverse-square law.



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Numerical optimization
(constrained)



They seek vowel coordinates x_i, y_i , for all i , that **minimize** the energy **E** of a vowel system, where that quantity is borrowed from physics and obeys an inverse-square law.

$$E = \sum_{i=1}^{n-1} \sum_{j=0}^{i-1} 1/r_{ij}^2 \quad \text{where } r_{ij}^2 = (x_i - x_j)^2 + (y_i - y_j)^2.$$

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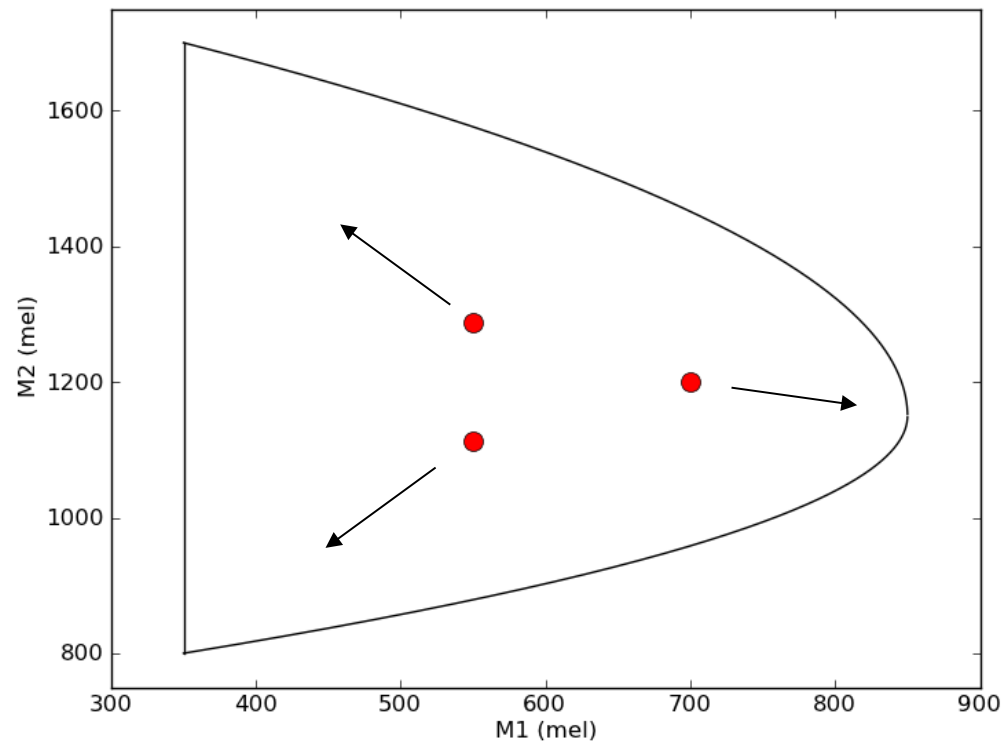
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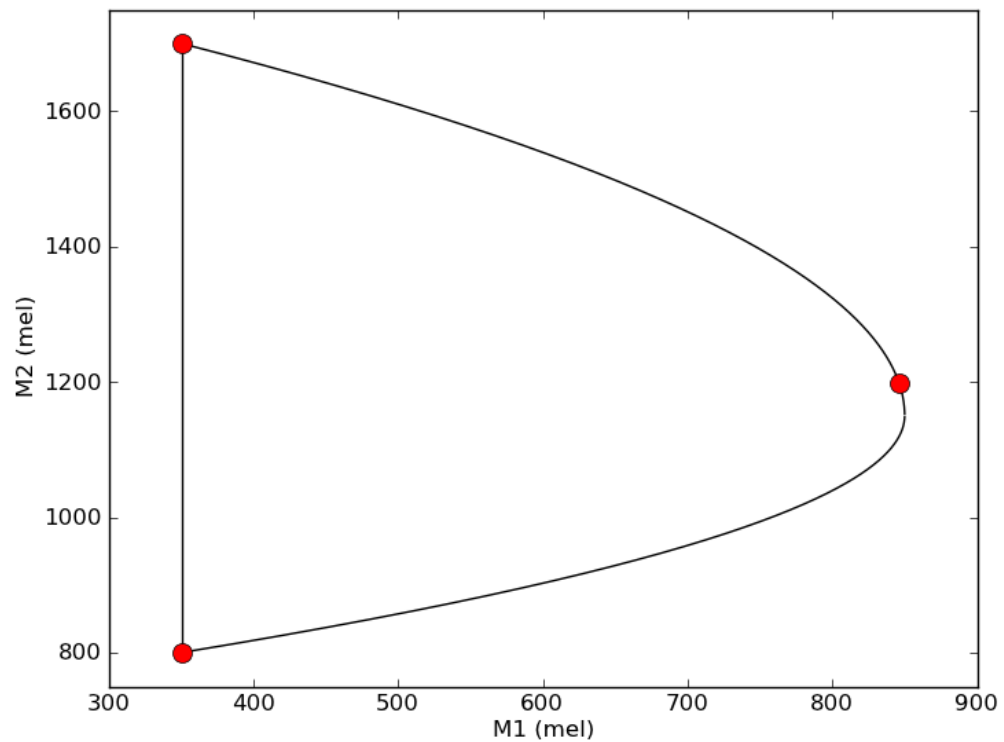
	0	1	2
0	r_{00}	r_{01}	r_{02}
1	r_{10}	r_{11}	r_{12}
2	r_{20}	r_{21}	r_{22}

$$\mathbf{E} = \sum_{i=1}^{n-1} \sum_{j=0}^{i-1} 1/r_{ij}^2$$

where $r_{ij}^2 = (x_i - x_j)^2 + (y_i - y_j)^2$.

	0	1	2
0	r_{00}	r_{01}	r_{02}
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Methods

- Simple hill-climbing procedure.

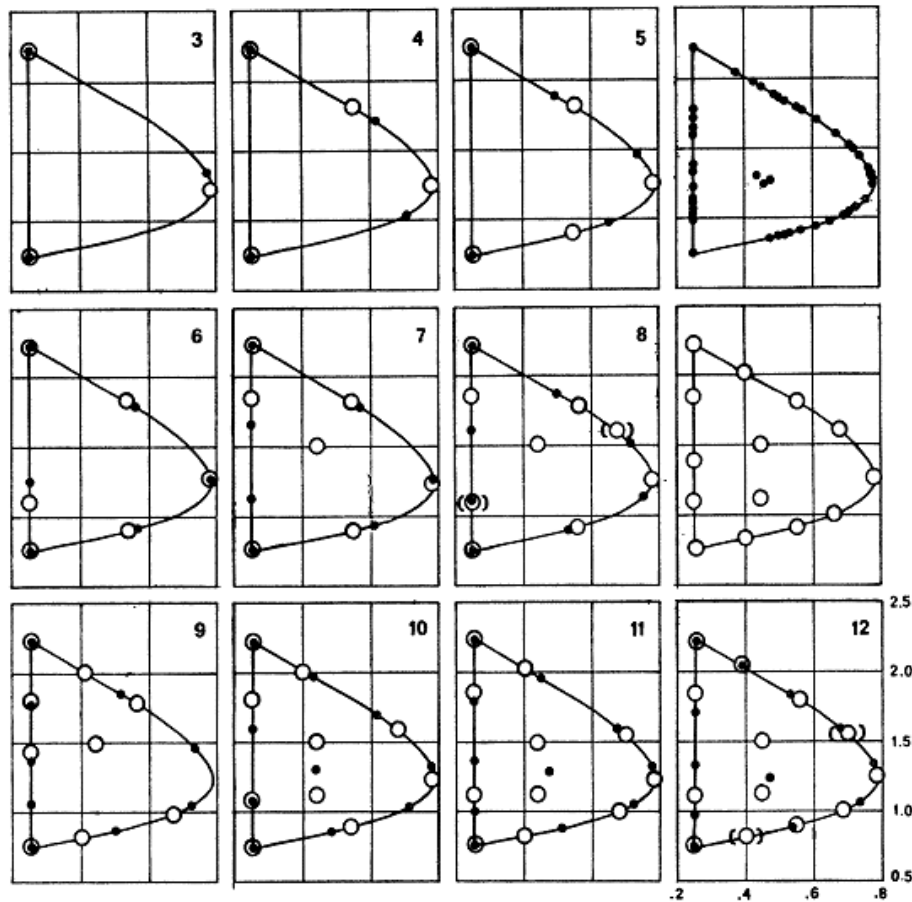
Methods

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Methods

- Simple hill-climbing procedure.
- “[T]he computer was left **overnight** to compute the solutions for $n = 3$ to $n = 12$ ”.
- Linguistic data: they use symbolic transcriptions for certain points in the space, and use these to plot vowel inventories from the literature.

SECOND FORMANT (kHz)



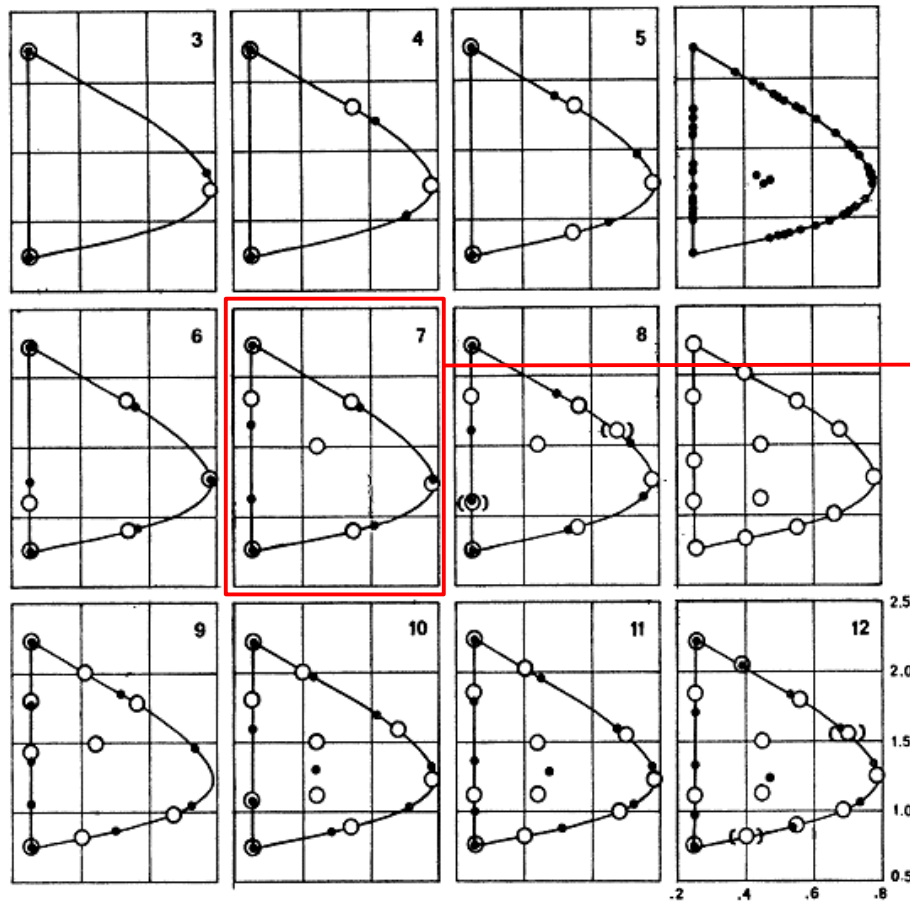
Data ○

Model ●

FIRST FORMANT (kHz)

FIGURE 4

SECOND FORMANT (kHz)



Data ○

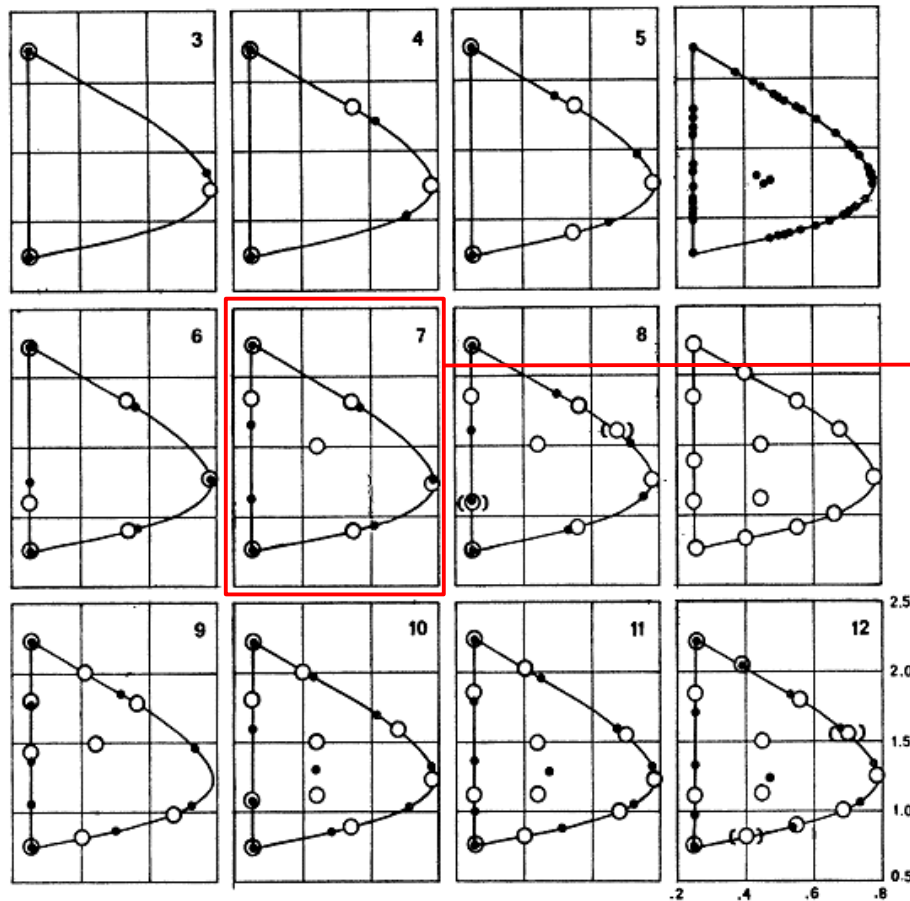
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Mismatches begin here, in systems with **central vowels**.

FIRST FORMANT (kHz)

FIGURE 4

SECOND FORMANT (kHz)



FIRST FORMANT (kHz)

FIGURE 4

Data ○

Model ●

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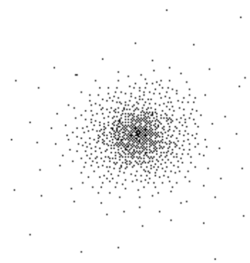
Model is focused on the **listener**, not the speaker.

Observations

- Inventory data are **coarse**, because taken from literature – could re-do more finely today.

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- Inventory data are **coarse**, because taken from literature – could re-do more finely today.
- “The question of uniqueness”: They produce only one point for each vowel, but implicitly there is a distribution: **cluster + outliers!**
- Tension between **listener** (modeled) and speaker (not).

Observations

- **Theoretical positioning:** “The phonetic mechanisms, as well as the directions for optimized use, must clearly ‘be assumed to be available to the child learning a language as an **a-priori, innate endowment**’ (Chomsky & Halle 1968:4)” – i.e. relevant to the generative tradition, but in an unexpected way.

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 - Glossed over: **how** the problem might be solved in reality – e.g. gradual historical change – this is **outside the scope**.